

Name: _____

Similar or not? If yes, name the postulate/theorem and the scale factor where possible.

1. Two triangles each have angles measuring 71° and 48° .
2. Sides 6, 10, 14 and 9, 15, 21.
3. Sides 12 and 18 with a 41° included angle; sides 8 and 12 with a 41° included angle.
4. One triangle has angles 55° and 72° ; another has angles 55° and 75° .
5. Sides 9, 12, 18 and 12, 16, 20.
6. Two segments cross, forming vertical angles. The pieces on one side are 8 and 6; the matching pieces on the other are 12 and 9.

Solve.

7. In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$ with D on $\overline{AB}$ and E on $\overline{AC}$. $AD = 10$, $DB = 6$, $AE = 15$. Find EC and AC.
8. $\triangle PQR \sim \triangle TUV$, $PQ = 20$, $TU = 25$, $QR = 4x$, $UV = 30$. Find x and QR.
9. A 6-ft-tall student casts an 8-ft shadow while a tree casts a 44-ft shadow. How tall is the tree?
10. Why does AA Similarity need only two pairs of angles when SSS needs all three pairs of sides?

